

Rotating Away the Location Degeneracy: an Exact Reparameterization for Hamiltonian Monte Carlo in Hierarchical and Joint Models

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Abstract

In a regression model with subject-level random effects, the likelihood identifies the sum of a subject-constant fixed effect and the matching combination of random effects, but not how that sum is split between them. Under the non-centred parameterization that Hamiltonian Monte Carlo (HMC) software uses by default, only the prior pins the split, and the result is a narrow ridge that slows the sampler on exactly the coefficients an analyst reports. We characterize the random-effect directions that fixed effects can absorb, for any design, give a construction that does not depend on how the design matrix happens to be parameterized, and state the condition a survival submodel adds in joint longitudinal-survival models. One fixed orthogonal rotation of the standardized random effects, the same for every random-effect column, turns these directions into a small block of sampling coordinates. It is exact for every value of the random-effect covariance: prior, posterior and every reported quantity are unchanged. A dense mass-matrix block over this block and the fixed effects then removes the ridge. In simulation the rotation raised the effective sample size per second of the intercept and the time slope by factors of 14 and 21 (geometric means); on two standard joint-model datasets the intercept and subject-constant covariates gained 17–28 times; and the same coefficients gained 5–7 times per gradient evaluation in a linear and a logistic mixed model outside the joint-model setting. A second finding concerns the association parameter of a joint model. For an ideal two-block sampler that draws it exactly from its conditional distribution given the other parameters, the lag-1 autocorrelation equals the fraction of its posterior variance those parameters explain. On real data that fraction is at least 0.85–0.87, mostly shared with the baseline hazard, which caps such a sampler at one effective draw in 13–15 and accounts for about three quarters of the measured gap between HMC and a Metropolis-within-Gibbs implementation.

Keywords: Hamiltonian Monte Carlo; non-centred parameterization; identifiability; preconditioning; mixed models; joint models.

1 Introduction

Mixed models with subject-level random effects are fitted by Hamiltonian Monte Carlo (HMC) and its adaptive variant, the No-U-Turn sampler (NUTS; Hoffman and Gelman, 2014), in general-

purpose software such as Stan (Carpenter et al., 2017), brms (Bürkner, 2017), rstanarm and NumPyro (Phan et al., 2019). brms and rstanarm generate the non-centred parameterization of the random effects by default, and it is the standard recommendation for models written directly in Stan or NumPyro: $b = B_{\text{std}}L^\top$ with B_{std} standard normal and L a Cholesky factor of the random-effect covariance. This removes the funnel between the random effects and their scale (Neal, 2003; Betancourt and Girolami, 2015) and is the reason these samplers work well on hierarchical models at all.

It leaves a second, simpler problem in place. Suppose the fixed-effects design contains a column that is constant within subject: the intercept, a treatment arm, sex. Adding a constant c to every subject’s random intercept and subtracting c from the fixed intercept leaves every fitted value unchanged. The likelihood identifies the sum; only the prior on the random effects, at scale $\sigma_0/\sqrt{N}$ for N subjects, pins the split. The same holds for every subject-constant covariate, whose coefficient trades off against a covariate-weighted combination of the random intercepts, and, whenever there is a random slope on time, for the fixed time slope and the mean random slope. In HMC’s sampling coordinates these directions are spread across all N subjects, the posterior has a ridge along them, and the sampler mixes slowly on the intercept, the subject-constant coefficients and the time slope: the coefficients an analyst reports. In a joint model of `pbc2`, for instance, the posterior correlation between the fixed intercept and the mean random intercept was -0.97 (Section 2.2).

The problem is old and well understood for Gibbs samplers. Hierarchical centring (Gelfand et al., 1995) removes it by sampling the centred effects, at the price of the funnel that non-centring was introduced to avoid, and the choice between the two is the subject of a large literature (Paspiliopoulos et al., 2003, 2007; Yu and Meng, 2011). A second route removes the directions: sum-to-zero constraints, whose effect on Gibbs and on NUTS Zanella and Roberts (2021) quantify; Stan’s `sum_to_zero_vector`; restricted spatial regression (Reich et al., 2006; Hodges and Reich, 2010); and, exactly, the transformation of Vines et al. (1996), which integrates the grand-mean directions out. The covariate-weighted direction for a random intercept appears in the discussion of Zanella and Roberts (2021) by Yang and Liu (2021) as a device for analysing Gibbs samplers, and Browne et al. (2009) name the problem directly, suggesting as future work that predictors be made orthogonal to the cluster indicators. Section 9.1 places the present work among these.

This paper does something different. It keeps the absorbable directions, and makes them sampling coordinates. We construct an orthonormal basis for every direction the fixed effects can absorb, in every random-effect column, and apply one fixed rotation to the standardized random effects so that these directions become a small block U of $k \times q$ coordinates, k being the number of directions and q the number of random-effect columns. A single dense block of the HMC mass matrix over (β, U) then whitens the ridge. The rotation acts on the subject index and the covariance factor on the column index, so they commute: the construction is exact for every value of the random-effect covariance, with no constraint, no correction and no change to any reported quantity.

The motivating application is the Bayesian joint model of a longitudinal marker and a time-to-event outcome (Wulfsohn and Tsiatis, 1997; Henderson et al., 2000; Tsiatis and Davidian, 2004;

Rizopoulos, 2012), as implemented by the R package `jmx`, which samples with NUTS in NumPyro. There the random effects also enter the hazard through the current marker value, and the survival submodel adds a condition on which directions remain absorbable. Joint models also raise a second question. Their primary estimand is the association parameter α , the log hazard ratio per unit of the current marker value. On it, NUTS was 3 to 8 times more efficient than the Metropolis-within-Gibbs sampler of JMbays2 (Rizopoulos et al., 2024) with no reparameterization at all. We show why: for an ideal two-block sampler that draws α exactly given the rest, the lag-1 autocorrelation equals the share of α ’s posterior variance the rest explains (Liu et al., 1994), and in joint models that share is large for a structural reason, the trade-off between α and the level of the log baseline hazard.

Contributions.

1. *The absorbable directions.* An exact characterization of the random-effect directions fixed effects can absorb, for any design and every random-effect column (Proposition 1); a construction that depends only on the column space of the design, with an example where the natural column-by-column check silently misses the slope direction (Example 1); and, for joint models, the condition the survival submodel imposes (Condition S, Proposition 2).
2. *An exact rotation and a small dense metric block.* One Householder rotation of the standardized random effects, costing $O(Nkq)$ per gradient, localizes every absorbable direction in a $(p + kq)$ -dimensional block (Proposition 3, Lemma 1). An analysis of the resulting ridge (Proposition 4) explains why a diagonal metric cannot handle it, why one dense block can, and where a fixed metric is weakest; a stress grid aimed at those weak spots kept the gains at 5–30 times.
3. *Blocked samplers and the association parameter.* The application of the data-augmentation autocorrelation identity to α (Proposition 5), with a bound that does not need Gaussianity, the measurement of the relevant R^2 on real data (0.85–0.87), and its decomposition, which places most of it in the survival block rather than the random effects.
4. *Evidence.* Simulation grids for one and two random-effect columns, a 100-replicate calibration study, two linear and generalized linear mixed models outside the joint-model setting, and two standard joint-model datasets, including a comparison with JMbays2.

Section 2 sets out the model and the degeneracy, Section 3 the absorbable directions, Section 4 the rotation and the metric, and Section 5 the blocked-sampler result. Sections 6–8 give the evidence and Section 9 discusses related work, costs and extensions. Proofs not given in the text are in Appendix A; the scripts that reproduce every table and figure are listed in Appendix B.

2 Model and the location degeneracy

2.1 Mixed models and joint models

Index subjects by $i = 1, \dots, N$ and subject i 's longitudinal observations by $j = 1, \dots, n_i$, taken at times t_{ij} . Subject i has a fixed-effects design $X_i \in \mathbb{R}^{n_i \times p}$ and a random-effects design $Z_i \in \mathbb{R}^{n_i \times q}$, and the linear predictor is

$$\eta_{ij} = x_{ij}^\top \beta + z_{ij}^\top b_i, \quad b_i \sim \mathcal{N}_q(0, \Sigma), \quad \Sigma = LL^\top, \quad (1)$$

with $L = \text{diag}(\sigma_b) L_{\text{corr}}$ lower triangular; for a random intercept and slope, $\sigma_b = (\sigma_0, \sigma_1)$ and ρ is their correlation. The response is Gaussian with mean η_{ij} and variance σ_e^2 , or follows a generalized linear model with linear predictor η_{ij} . Collect the random effects as the rows of $B \in \mathbb{R}^{N \times q}$. The non-centred parameterization samples

$$B_{\text{std}} \in \mathbb{R}^{N \times q}, \quad \text{entries iid } \mathcal{N}(0, 1), \quad B = B_{\text{std}} L^\top, \quad (2)$$

and θ collects every other parameter, including β , σ_b and L_{corr} .

In a joint longitudinal-survival model the longitudinal submodel is Gaussian, and subject i 's current underlying marker value

$$m_i(t) = x_i(t)^\top \beta + z_i(t)^\top b_i \quad (3)$$

enters a proportional hazards model,

$$h_i(t) = h_0(t) \exp\{\gamma^\top w_i + \alpha m_i(t)\}, \quad (4)$$

with baseline covariates w_i and association parameter α . Here $x_i(t)$ and $z_i(t)$ are the longitudinal design rows evaluated at time t with subject i 's baseline covariates. In `jmjax` the log baseline hazard is a cubic B-spline with a second-order random-walk smoothing prior, and the cumulative hazard is computed by Gauss–Legendre quadrature, so the random effects reach the likelihood through η_{ij} and through $m_i(t)$ at each event time and each quadrature node. The priors on β , γ and α are normal, the random-effect standard deviations have gamma or half-normal priors, and L_{corr} an LKJ prior; none of what follows depends on these choices beyond the prior on B_{std} being iid standard normal.

2.2 The degeneracy

Take a random intercept and slope, $z_{ij} = (1, t_{ij})^\top$, and a fixed-effects design with an intercept, a time column and subject-constant covariates such as age and sex. Three kinds of shift leave every η_{ij} and every $m_i(t)$ unchanged:

- $b_{i0} \mapsto b_{i0} + c$ for all i , with $\beta_0 \mapsto \beta_0 - c$;

- $b_{i0} \mapsto b_{i0} + c \text{age}_i$ for all i , with $\beta_{\text{age}} \mapsto \beta_{\text{age}} - c$, and likewise for every subject-constant covariate;
- $b_{i1} \mapsto b_{i1} + c$ for all i , with $\beta_{\text{time}} \mapsto \beta_{\text{time}} - c$.

The second kind adds one direction for each subject-constant covariate. The third is present whenever there is a random slope, whatever the covariates. Along each shift the likelihood is exactly flat, and only the prior on the random effects locates the split, at scale $\sigma_0/\sqrt{N}$ for the mean intercept direction.

In the coordinates HMC samples, a shift of this kind moves all N entries of one or both columns of B_{std} together (for the mean intercept direction with uncorrelated random effects, each by c/σ_0), against one or a few coordinates of β . A diagonal mass matrix, which is what NUTS adapts by default, can only rescale coordinates one at a time and cannot align with such a direction. In a fit of the `pbc2` data (Section 8) with an intercept and one subject-constant covariate, the posterior correlation between β_0 and the mean random intercept was -0.971 , and between the covariate's coefficient and the matching covariate-weighted combination -0.972 . The two affected coefficients had minimum effective sample sizes of 441 and 526 from 4000 draws, against 2893 and more than 4000 for the identified sums. Stan's user's guide documents the same mechanism for a model with two intercepts under weak priors (Stan Development Team, 2025).

The degeneracy is not a funnel. The funnel is a problem of curvature that changes with a scale parameter; here the curvature along the ridge is nearly constant and the problem is that the ridge is narrow in a direction the metric cannot see. It is also not the non-identifiability of an improper model: the prior on the random effects makes the posterior proper and the split identified, weakly.

3 Absorbable directions

The rotation of Section 4 is exact for any fixed orthogonal matrix. What decides whether it does anything useful is the basis it is built from. The basis should contain every direction the fixed effects can absorb, so that none is left spread across the N subject coordinates, and only such directions, so that the dense block is not spent on directions the data pin down. This section characterizes them.

Fix a random-effect column c and write $z_i = Z_i e_c \in \mathbb{R}^{n_i}$ for subject i 's entries of it; in this section only, $z_i(t)$ denotes the scalar entry of column c in the design row at time t , rather than the whole row of (3).

Definition 1 (Absorbable direction). A vector $v \in \mathbb{R}^N$ is absorbable for column c if there is a witness $d \in \mathbb{R}^p$ such that the shift $Be_c \mapsto Be_c + v$, $\beta \mapsto \beta + d$ leaves every linear predictor and every marker trajectory, and hence the likelihood, unchanged:

$$X_i d + v_i z_i = 0 \quad \text{for every } i, \tag{L}$$

$$x_i(t)^\top d + v_i z_i(t) = 0 \quad \text{for every } i \text{ and every } t \in [0, \tau]. \tag{T}$$

Write $\mathcal{A}_c$ for the set of absorbable directions and $\mathcal{A}_c^{\text{long}}$ for the set satisfying (L) alone, which is the whole requirement in a mixed model without a survival submodel.

Both sets are linear subspaces of $\mathbb{R}^N$. The likelihood uses $m_i(t)$ only up to each subject's event or censoring time, so requiring (T) on all of $[0, \tau]$ is slightly more than invariance of the likelihood needs; it is what makes the set of directions a property of the model rather than of the observed follow-up. The definition concerns the likelihood only: the prior on the random effects is not invariant under these shifts, and that asymmetry is what identifies the split at all.

3.1 Characterization and construction

Let $\mathcal{T} : \mathbb{R}^p \rightarrow \bigoplus_i \mathbb{R}^{n_i}$, $(\mathcal{T}d)_i = X_i d$, map a shift of β to its effect on the stacked linear predictor, and $\mathcal{U}_c : \mathbb{R}^N \rightarrow \bigoplus_i \mathbb{R}^{n_i}$, $(\mathcal{U}_c v)_i = v_i z_i$, do the same for a shift of random-effect column c . Let $P_{\mathcal{T}}$ be the orthogonal projector onto $\text{range } \mathcal{T}$ and, for $z_i \neq 0$, P_{z_i} the projector onto $\text{span}(z_i)$. Let $S_c = \{i : z_i \neq 0\}$ be the subjects with at least one observation at which column c is nonzero, and let $M_c \in \mathbb{R}^{(\sum_i n_i) \times p}$ stack $(I - P_{z_i})X_i$ for $i \in S_c$ and X_i for $i \notin S_c$.

Proposition 1 (Characterization). (a) $\mathcal{A}_c^{\text{long}} = \mathcal{U}_c^{-1}(\text{range } \mathcal{T}) = \ker\{(I - P_{\mathcal{T}})\mathcal{U}_c\}$. It depends on the fixed-effects design only through its column space.

(b) A vector d witnesses some $v \in \mathcal{A}_c^{\text{long}}$ if and only if $d \in \ker M_c$. The witnessed v then has $v_i = -z_i^\top X_i d / z_i^\top z_i$ for $i \in S_c$, and v_i is unrestricted for $i \notin S_c$.

Proof. (a) Condition (L) says $\mathcal{T}d = -\mathcal{U}_c v$ for some d , that is, $\mathcal{U}_c v \in \text{range } \mathcal{T}$, which is equivalent to $(I - P_{\mathcal{T}})\mathcal{U}_c v = 0$. (b) For $i \in S_c$, $X_i d = -v_i z_i$ holds for some scalar v_i if and only if $X_i d \in \text{span}(z_i)$, that is, $(I - P_{z_i})X_i d = 0$, and then $v_i = -z_i^\top X_i d / z_i^\top z_i$. For $i \notin S_c$, (L) reads $X_i d = 0$ whatever v_i is. $\square$

Part (b) is the construction used in practice. In a mixed model without a survival submodel, a subject with $z_i = 0$ has no random effect in column c that enters the likelihood, so we may take S_c to be every subject. Then one singular value decomposition of the $(\sum_i n_i) \times p$ matrix M_c gives a basis of $\ker M_c$, each basis vector d gives one direction $v(d)$, and, if X has full column rank, $d \mapsto v(d)$ is injective, so $\dim \mathcal{A}_c^{\text{long}} = \dim \ker M_c \leq p$. The computation is done once, before sampling. In the implementation every returned direction is also checked against (L) before use, so a loose rank tolerance cannot admit a direction that is not absorbable.

3.2 The survival submodel

In a joint model the requirement (T) quantifies over all $t \in [0, \tau]$, while M_c sees only the observation times.

Condition S (Structural extension). For every $d \in \ker M_c$ and every subject i there is a scalar $v_i(d)$, equal to $-z_i^\top X_i d / z_i^\top z_i$ when $i \in S_c$, such that $x_i(t)^\top d + v_i(d) z_i(t) = 0$ for every $t \in [0, \tau]$. (Under the assumption of Proposition 2, $v_i(d)$ is then unique for every i .)

Proposition 2 (Joint model). *Suppose every subject $i \notin S_c$ has some $t \in [0, \tau]$ with $z_i(t) \neq 0$. Under Condition S, $\mathcal{A}_c = \{v(d) : d \in \ker M_c\}$. For a subject $i \notin S_c$, $v_i(d)$ is determined by (T) at any t with $z_i(t) \neq 0$; in particular it is not free, although the longitudinal data alone leave it free.*

The proof is in Appendix A. Condition S holds for every column type that commonly arises: a subject-constant column $(x_i(t))^\top e_j \equiv \text{const}$, absorbed by the intercept column $z_i(t) \equiv 1$, the time column $(x_i(t))^\top e_j = t$, absorbed by a random slope $z_i(t) = t$ and a covariate-by-time interaction $(c_i t)$. In each the identity holds as a function of t , not by coincidence at the observed times. More generally it holds whenever the functions involved lie in a space determined by their values at a subject's observed times, such as polynomials of degree below n_i . It can fail for subjects with few observations relative to the flexibility of the time basis. It is also cheap to check: the design rows at the event times and quadrature nodes are needed for the likelihood anyway, and `jmjax` evaluates the identity there at every fit and warns if the worst relative residual exceeds 10^{-6} . Proposition 2 also settles how to treat a subject with no observation at which the column is nonzero, such as a baseline-only subject for the slope column: its entry is forced by the survival submodel, not free, and setting it to zero would produce a direction the likelihood does identify.

3.3 A design where the natural check fails

A natural construction, and the one `jmjax` originally used, looks for columns of X that are *individually* subject-constant multiples of z : column j is kept if $X_i e_j = s_i z_i$ for every i , and the basis is the span of the resulting vectors s . This finds a subspace of $\mathcal{A}_c^{\text{long}}$, but not always all of it.

Example 1. Let every subject be observed at the same $m \geq 3$ distinct times, let $Z_i = [1, t]$ with c the slope column, and let $X_i = [1, t + \frac{1}{2}t^2, t^2]$. The ratios of the three columns to t are $1/t$, $1 + t/2$ and t , none constant, so the column-by-column search returns nothing. Yet $d = (0, 1, -\frac{1}{2})$ gives $X_i d = t$, so $v = -\mathbf{1}_N$ satisfies (L) and (T), and $\mathcal{A}_c \neq \{0\}$.

The column space in Example 1 is $\text{span}\{1, t, t^2\}$, exactly what an orthogonal-polynomial or natural-spline mean in time produces; the monomial spelling `time + I(time^2)` is caught, because there the time column appears on its own. So with a spline mean and a random slope, the column-by-column search silently leaves the slope degeneracy in place, while still reporting a basis for the intercept column. By Proposition 1(a) the absorbable set depends only on the column space, and the null-space construction of Proposition 1(b) is invariant to how that space is spanned. It is what `jmjax` now uses; the column-wise search is retained only to report which covariates expose each direction.

Finally, let Q_c be an orthonormal basis of $\mathcal{A}_c$ and $Q \in \mathbb{R}^{N \times k}$ an orthonormal basis of $\text{span} \bigcup_c \mathcal{A}_c$. In the simulation design of Section 6 (an intercept, time, age and sex, with a random intercept and slope) $\mathcal{A}_0 = \text{span}\{\mathbf{1}, \text{age}, \text{sex}\}$ and $\mathcal{A}_1 = \text{span}\{\mathbf{1}\}$, so $k = 3$.

4 The rotation and the metric

4.1 Construction and exactness

Complete Q to an orthogonal matrix $R = [Q \mid Q_\perp] \in \mathbb{R}^{N \times N}$. We use the Householder completion: R is stored as k reflectors, so applying R or $R^\top$ to a vector costs $O(Nk)$ and no $N \times N$ matrix is formed. Instead of B_{std} , sample

$$B_{\text{std}} = R \begin{bmatrix} U \\ V \end{bmatrix} = QU + Q_\perp V, \quad B = B_{\text{std}} L^\top, \quad (5)$$

where $U \in \mathbb{R}^{k \times q}$ and $V \in \mathbb{R}^{(N-k) \times q}$ have iid $\mathcal{N}(0, 1)$ entries.

Proposition 3 (Exactness). *For any fixed orthogonal R : (i) the map $(U, V) \mapsto B_{\text{std}}$ is linear and volume-preserving; (ii) the prior it induces on B_{std} is the original iid $\mathcal{N}(0, 1)$ prior; (iii) the posterior of (B, θ) is identical to that of the non-centred parameterization (2), for every value of L ; and (iv) $U = Q^\top B_{\text{std}}$.*

Proof. $\text{vec}(B_{\text{std}}) = (I_q \otimes R) \text{vec}([U; V])$ and $I_q \otimes R$ is orthogonal, which gives (i), and (ii) because $(I_q \otimes R) \mathcal{N}(0, I_{Nq}) = \mathcal{N}(0, I_{Nq})$. The likelihood depends on (B_{std}, θ) only, and their joint prior is unchanged, so the posterior pushed forward to (B, θ) is unchanged: (iii). (iv) follows from $Q^\top Q_\perp = 0$. $\square$

Two choices in (5) matter. First, the rotation acts on the sampled, standardized coordinates. Rotating the draws after sampling would change nothing about the sampler's geometry, and sampling rotated *centred* effects $R^\top B$ would reintroduce the dependence on σ_b that non-centring removes. Second, R acts on the subject index, the same for every column, while L acts on the column index. The two commute, $R^\top B = (R^\top B_{\text{std}}) L^\top$, which is why the construction is exact for every value of the covariance with no correction. A rotation of the stacked Nq -vector designed to align particular intercept-slope combinations with coordinate axes would also be exact, but those combinations depend on L , so a fixed rotation of that kind would align them only at one value of the correlation.

For a random intercept alone ($q = 1$), `jmjax` samples the centred effects $b \sim \mathcal{N}(0, \sigma_b^2 I_N)$ directly, and the rotation is applied to b . Because the covariance is a multiple of the identity, $R^\top b$ has the same prior for every σ_b and Proposition 3 holds unchanged.

4.2 Where the degeneracy goes

Lemma 1 (Localization). *Let $v \in \mathcal{A}_c$ with witness d . In the coordinates (β, U, V) , the likelihood-invariant shift $(\beta, B e_c) \mapsto (\beta + d, B e_c + v)$ is*

$$\beta \mapsto \beta + d, \quad U \mapsto U + (Q^\top v) e_c^\top (L^\top)^{-1}, \quad V \mapsto V,$$

for every value of L . Every such shift therefore lies in the coordinate block (β, U) , of dimension $p + kq$.

Proof. Adding v to column c of $B = B_{\text{std}} L^\top$ adds $\Delta = v e_c^\top (L^\top)^{-1}$ to B_{std} . Every column of Δ is a multiple of $v \in \text{span}(Q)$, so $Q_\perp^\top \Delta = 0$ and $Q^\top \Delta = (Q^\top v) e_c^\top (L^\top)^{-1}$. $\square$

Without the rotation, the same shift has a component on every one of the N rows of B_{std} . With it, the flat directions of the likelihood mix the intercept and slope columns of U in proportions set by the sampled L , but always stay inside the fixed block (β, U) . This is why every column is rotated by the same Q : rotating the intercept column alone captures the directions only when the random effects are uncorrelated.

4.3 The ridge, and the dense block

Being flat in the likelihood is not the same as being flat in the posterior: along each invariant direction the priors on β and B_{std} still locate the split. The next result describes the ridge this leaves in one plane.

Proposition 4 (The ridge). *Consider one invariant direction in the reduced model that holds every other coordinate fixed. Let $\tilde{\beta}$ be the coordinate of β along the witness, with a flat prior, and u the matching coordinate of U , with prior $\mathcal{N}(0, 1)$, and suppose the likelihood informs only $s = \tilde{\beta} + \kappa u$, with precision τ . Then*

- (i) $\text{corr}(\tilde{\beta}, u \mid y) = -\{1 + 1/(\tau\kappa^2)\}^{-1/2}$;
- (ii) *under a diagonal metric, the whitened covariance of the plane has eigenvalues $1 \pm |r|$, r the correlation in (i): it is never slow along the ridge, but has a narrow direction of variance $1 - |r| \approx 1/(2\tau\kappa^2)$, which caps the step size;*
- (iii) *a dense metric over $(\tilde{\beta}, u)$ whitens the plane exactly;*
- (iv) *for the mean direction of the random intercepts, $\kappa = \sigma_0/\sqrt{N}$ and $\tau = \sum_i n_i/\sigma_e^2$, so $\tau\kappa^2 = \bar{n}\sigma_0^2/\sigma_e^2$, the information each subject carries about its own level. It does not depend on N .*

Proof. The posterior precision of $(\tilde{\beta}, u)$ is $\tau(1, \kappa)^\top(1, \kappa) + \text{diag}(0, 1)$, whose inverse is

$$\tau^{-1} \begin{pmatrix} 1 + \tau\kappa^2 & -\tau\kappa \\ -\tau\kappa & \tau \end{pmatrix};$$

this gives (i). Whitening a 2×2 covariance by its diagonal gives its correlation matrix, whose eigenvalues are $1 \pm |r|$: (ii). (iii) is the definition of a dense metric. For (iv), moving the mean direction by u moves every subject's level by $\sigma_0 u/\sqrt{N}$, and the data inform a common shift of all levels with precision $\sum_i n_i/\sigma_e^2$. $\square$

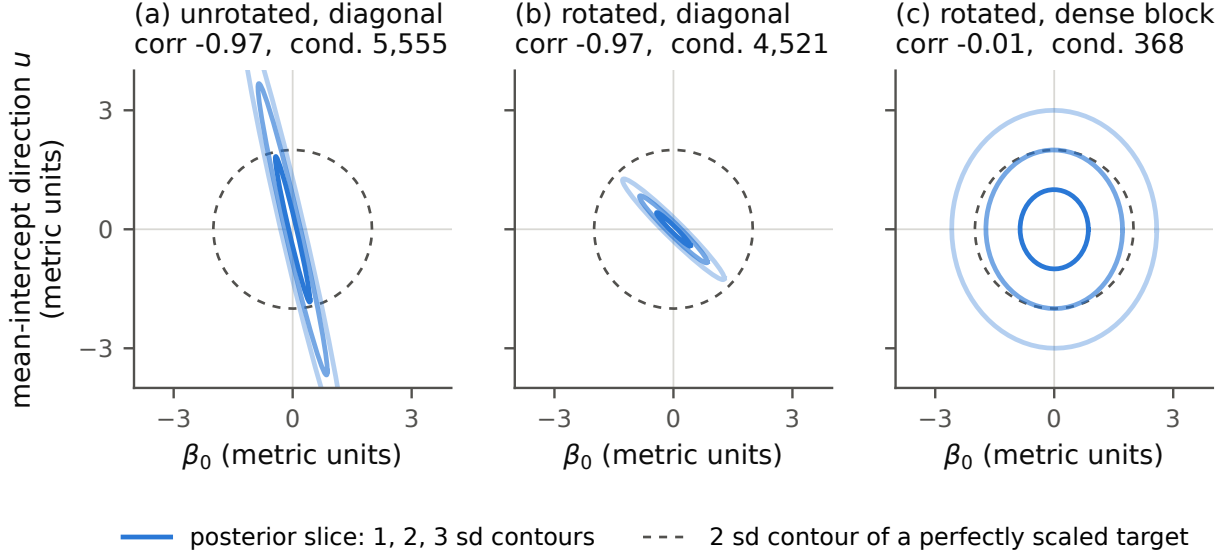


Figure 1: The location ridge and what removes it. Exact conditional posterior of the fixed intercept β_0 and the mean intercept direction u , through the mode, in the units the HMC metric sees them (Gaussian longitudinal submodel, variance components fixed; $N = 300$, 2–11 visits, $\rho = 0.3$, $\sigma_e = 0.3$). Solid: 1, 2 and 3 standard deviation contours; dashed: the 2 standard deviation contour of a perfectly scaled target. (a) Unrotated coordinates, diagonal metric: u is spread over all N random intercepts and trades off almost one for one with β_0 . (b) The rotation makes u a single coordinate but leaves the ridge. (c) The dense metric block over (β, U) removes it. “cond.” is the condition number of the whole whitened posterior.

The statement is exact for the mean intercept direction, whose witness is the single coordinate β_0 . For other directions the witness mixes several coordinates of β , and (ii)–(iii) then describe the whole (β, U) block. The rotation alone therefore trades a direction spread over N coordinates for one concentrated in a few, but under a diagonal metric that direction is still narrow, and narrowest where the degeneracy is worst: many visits per subject, or little noise. A dense metric block over (β, U) removes it. Without the rotation, the same fix would need a dense metric over all Nq random-effect coordinates, too large to adapt reliably during warm-up and $O(N^2q^2)$ per gradient.

Figure 1 shows this in the exact Gaussian posterior of the longitudinal submodel with the variance components fixed at the simulation values of Section 6 ($N = 300$, 2–11 visits, $\rho = 0.3$, $\sigma_e = 0.3$). The conditional correlation of β_0 and the mean intercept direction is -0.97 in the unrotated coordinates and still -0.97 after rotation with a diagonal metric; the dense block brings it to -0.01 . The condition number of the whole whitened posterior falls from 4,521 to 368. Across the designs examined, the dense block cut it by factors of 5 to 70, for example from 12,037 to 174 with 8–20 visits per subject; what remains is conditioning any fit has, such as the correlation between each subject’s own intercept and slope.

4.4 What a fixed metric cannot do

By Lemma 1, the ridge’s orientation depends on the sampled L : on σ_0 through $\kappa \propto \sigma_0$, and, for directions shared between the intercept and slope columns, on the correlation ρ . NUTS adapts one fixed metric. For the σ_0 part the loss can be bounded. Let δ be the posterior coefficient of variation of σ_0 , roughly $1/\sqrt{2N}$. A metric that whitens the plane exactly at one value of σ_0 leaves a condition number of $1 + O(\delta\sqrt{\tau\kappa^2})$ elsewhere in the posterior, that is,

$$1 + O\left(\sqrt{\bar{n}\sigma_0^2/(\sigma_e^2N)}\right),$$

which is small when there are many subjects relative to the information each carries, the usual regime for longitudinal data. In the exact Gaussian model of Figure 1 the worst block condition number over $\sigma_0(1 \pm \delta)$ was 1.76, 1.50 and 1.31 for $N = 150, 300$ and 600 with 2–11 visits, and 2.71, 2.05 and 1.64 with 8–20 visits. The analysis does not cover the ρ part, which grows as $|\rho| \rightarrow 1$. These are the two regimes Section 6.4 stress-tests. For $q = 1$ in centred coordinates, the ridge’s orientation does not involve σ_b ; only the scale of the rotated coordinate’s prior does, a mismatch of order δ rather than $\delta\sqrt{\tau\kappa^2}$.

4.5 Implementation and cost

`jmxax` implements (5) in NumPyro: U and V are the sampled sites, B_{std} is a deterministic function of them computed with q applications of the k reflectors, and the mass matrix is dense on (β, U) and diagonal elsewhere, adapted during warm-up in the usual way. The extra cost per gradient is $O(Nkq)$. In Section 6’s design the dense block has $p + kq = 10$ coordinates, and wall time per fit was 0.83–1.10 times that of the unrotated fit. The basis is computed once from the design matrices, before sampling. The rotation and the dense block are the package’s default for NUTS fits with a random intercept only, or with $q \geq 2$ correlated random effects.

5 Blocked samplers and the association parameter

The location degeneracy does not involve α . Yet on α , `jmxax`’s NUTS was 3–8 times more efficient per second than `JMbayes2` even without the rotation: 5.7–8.1 times across a replicated simulation sweep with 1 to 12 covariates, and 3.2 and 7.6 times on the two datasets of Section 8. `JMbayes2` updates the parameters in blocks, with adaptive random-walk Metropolis steps for the coefficients. This section shows that most of the difference is a property of blocked updating in joint models rather than of either implementation.

Write ψ for every parameter other than α : the random effects, β , γ , the baseline-hazard coefficients and the variance components.

Proposition 5 (Blocked updates of α). *Consider the two-block sampler that draws α exactly from*

$p(\alpha \mid \psi, y)$ and then ψ exactly from $p(\psi \mid \alpha, y)$, and let

$$r = \frac{\text{Var}\{E(\alpha \mid \psi, y)\}}{\text{Var}(\alpha \mid y)} = 1 - \frac{E\{\text{Var}(\alpha \mid \psi, y)\}}{\text{Var}(\alpha \mid y)},$$

the fraction of α 's posterior variance explained by the other parameters, and $R^2 \leq r$ the squared multiple correlation of α on ψ .

- (i) At stationarity the lag-1 autocorrelation of the α chain is r .
- (ii) Its integrated autocorrelation time is at least $(1+r)/(1-r)$, so its effective sample size per draw is at most $(1-r)/(1+r) \leq (1-R^2)/(1+R^2)$.
- (iii) If the posterior is Gaussian, $r = R^2$, the α chain is $AR(1)$ with coefficient R^2 , and the bound in (ii) is attained.

Proof. The α chain of a two-block sampler is a data-augmentation chain. Its lag-1 autocovariance is $\text{Var}\{E(\alpha \mid \psi, y)\}$ and its transition operator is self-adjoint and positive on L^2 of the posterior of α (Liu et al., 1994), which gives (i); the second expression for r is the law of total variance, and $R^2 \leq r$ because the best linear predictor of α explains no more variance than $E(\alpha \mid \psi, y)$. For (ii), the spectral theorem writes the lag- k autocorrelation of α as $\rho_k = \int \lambda^k \mu(d\lambda)$ for a probability measure μ on $[0, 1]$, so Jensen's inequality gives $\rho_k \geq \rho_1^k = r^k$ and $1 + 2 \sum_{k \geq 1} \rho_k \geq (1+r)/(1-r)$; the function $(1-x)/(1+x)$ is decreasing. For (iii), $E(\alpha \mid \psi)$ is linear in ψ and $E(\psi \mid \alpha)$ linear in α ; composing the two regressions gives $\alpha_{t+1} = c + R^2 \alpha_t + \varepsilon_t$ with ε_t independent of the past, since the coefficient is $\text{Cov}(\alpha, \psi) \Sigma_\psi^{-1} \text{Cov}(\psi, \alpha) / \text{Var}(\alpha) = R^2$, and then $\rho_k = (R^2)^k$. $\square$

Part (ii) is what makes a measured linear R^2 usable: it bounds the efficiency of any exact two-block sampler from above, whatever the shape of the posterior. The proposition is an application of known theory; the two-block sampler is a benchmark, not a model of JMBayes2. Finer blocking and random-walk Metropolis steps can be expected to make a real implementation slower still (Liu et al., 1994; Roberts and Sahu, 1997), though neither is guaranteed to in every case. What the proposition makes precise is the contrast with HMC, which moves α and ψ together. Under a fixed metric, a correlation between them is part of the posterior's shape and is paid for in leapfrog steps, which grow with the square root of the condition number (Neal, 2011). A single correlation at level R^2 contributes a condition number $(1+\sqrt{R^2})/(1-\sqrt{R^2})$, so the cost grows at most like $1/\sqrt{1-R^2}$, against $1/(1-R^2)$ for the blocked sampler. Often it grows not at all, because in a joint model the step size is already set by some other narrow direction, such as the spline coefficients or the variance components. In a Gaussian check, comparing a simulated two-block Gibbs sampler with a stylised model of HMC (exact Hamiltonian dynamics, random integration time), the two-block sampler's lag-1 autocorrelation matched R^2 to within 0.004, and its integrated autocorrelation time went 3.0, 9.0, 27.6, 39.0 and 199 as R^2 went 0.5, 0.8, 0.93, 0.95 and 0.99. The HMC model's stayed between 1.6 and 2.1, at a cost of 4.8 to 46.8 gradient evaluations per independent draw; with an unrelated narrow ridge already present, the step count was 22.2 at every R^2 from 0 to 0.99.

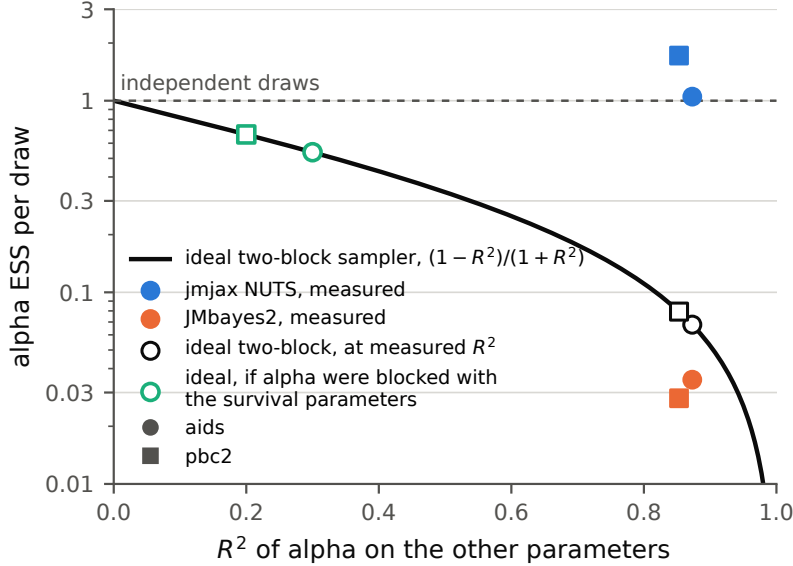


Figure 2: Effective sample size per draw for α . Curve: an ideal two-block sampler, $(1 - R^2)/(1 + R^2)$ (Proposition 5). Black open markers: the ideal sampler at the measured R^2 (0.873 for **aids**, 0.853 for **pbc2**). JMbayer2 does two to three times worse than that ideal; the NUTS draws are effectively independent. Green open markers: the ideal if α were updated in one block with the survival parameters, so that only the random effects are conditioned away ($R^2 = 0.30$ and 0.20).

5.1 How large R^2 is in a joint model

Two things make R^2 large. The first is errors in variables: given the random effects, each marker trajectory $m_i(t)$ is known and α is the coefficient of a known time-varying covariate, which the data pin sharply; marginally, α also carries the uncertainty in every trajectory. The second is the level of the hazard. Shifting $m_i(t)$ by a constant for every subject changes $\alpha m_i(t)$ by an amount the log baseline hazard absorbs, so α is correlated with the baseline-hazard level unless the marker is centred.

We estimated R^2 by regressing α on every other sampled quantity in a long rotated fit (4 chains of 5000 draws; 960 regressors for **aids** and 649 for **pbc2**, adjusted for their number), and split it by source. On **aids**, $R^2 = 0.873$; the survival block alone (γ and the baseline-hazard coefficients) explained 0.86 of α 's posterior variance and the random effects alone 0.30. On **pbc2** the figures were 0.853, 0.82 and 0.20. The hazard level, not the random effects, is what a blocked sampler mostly conditions away. By Proposition 5(ii), the linear R^2 is a lower bound on the fraction r , so the efficiency it implies for an ideal blocked sampler is an upper bound.

Figure 2 puts the measurements against the prediction. At the measured R^2 an ideal two-block sampler keeps at most 0.068 (**aids**) and 0.079 (**pbc2**) effective draws per draw, one in 13–15. JMbayer2 achieved 0.035 and 0.028, and jmjax's NUTS 1.05 and 1.72 in the same long fit. On a log scale, the ideal benchmark accounts for 80% of the measured ESS-per-draw gap on **aids** (15.5 of 30 times) and 75% on **pbc2** (21.7 of 61 times); the rest, a factor of two to three, is attributable

to the implementation. A blocked sampler could escape most of the structural part by updating α jointly with γ and the baseline-hazard coefficients: with only the random effects conditioned away, the ideal ESS per draw would be about 0.54 and 0.67. But that block has 10–20 strongly correlated dimensions without exact conditional draws, and an adaptive random-walk proposal on it is itself slow. NUTS obtains the joint move without any block design.

6 Simulation

6.1 Design

Data were simulated from a joint model with a random intercept and slope, longitudinal mean $\beta_0 + \beta_1 t + \beta_{\text{age}} \text{age}_i + \beta_{\text{sex}} \text{sex}_i$ with $\beta_0 = 2$ and $\beta_1 = 0.5$, $\text{age}_i \sim \mathcal{N}(0, 10^2)$ and $\text{sex}_i \sim \text{Bernoulli}(0.5)$, $\sigma_b = (0.8, 0.3)$, $\sigma_e = 0.3$, and a Weibull hazard (shape 1.3) with $\alpha = 0.5$ and a treatment effect $\gamma = -0.3$, calibrated to an event rate of about one half. Visits were scheduled at regular intervals up to $t = 10$ and stopped at the event or censoring time. All fits used the spline baseline hazard of Section 2.1, so the fitted model does not contain the true hazard exactly. Each design was simulated with three seeds, and every seed was fitted twice on the same data: unrotated, and with the rotation and its dense block. Both fits report the same parameters, β included, so the comparison is on the quantities a user reports. Effective sample size (ESS) and split $\hat{R}$ are the multi-chain estimators computed by NumPyro (Gelman et al., 2013); ESS per second uses each fit’s own sampling time, and ratios are rotated over unrotated, paired by seed, with the geometric mean over seeds reported for each design.

The core grid crossed about 9 against about 17 visits per subject with $\rho \in \{0, 0.3, 0.6, 0.9\}$ at $N = 300$, and added $N \in \{150, 600\}$ at 9 visits and $\rho = 0.3$: ten designs, 2 chains of 500 warm-up and 1000 sampling iterations.

6.2 Efficiency

Table 1 gives the results. Over all 30 paired fits the geometric-mean gain was 13.5 for the intercept (per seed 4.5–31.6) and 21.1 for the time slope (5.1–70.8). Wall time was essentially unchanged (ratio 1.03, range 0.83–1.10), so the gain is almost entirely in ESS per draw: under the rotation the median ESS over 2000 draws was about 3,450 for both coefficients, above the number of draws, against 180–235 unrotated. There were no divergences, and the worst $\hat{R}$ on a reported coefficient was 1.002 rotated and 1.044 unrotated.

The pattern follows Proposition 4. The gain is largest with dense visits, where each subject carries the most information about its own random effects (19–25 times for the intercept, 25–59 for the slope), and does not grow with N (7.6 at $N = 150$, 8.2 at $N = 600$), because the ridge is set by per-subject information, not by the number of subjects. Figure 3 shows all three grids against that information, $\bar{n} \sigma_0^2 / \sigma_e^2$.

The association parameter, which the degeneracy does not involve, is not untouched: its ratio split by design, 0.72–1.00 in the six designs with about 9 visits and 1.32–1.59 in the four with about

Table 1: Core grid: ESS per second for the intercept and the time slope (means over 3 seeds), and the rotated/unrotated ratio (geometric mean of the paired per-seed ratios).

visits	ρ	N	intercept			time slope			α
			unrot.	rotated	ratio	unrot.	rotated	ratio	ratio
9	0.0	300	25.9	354.4	14.2	9.1	304.9	36.4	0.85
17	0.0	300	12.1	229.4	19.3	4.0	234.9	59.2	1.32
9	0.3	300	30.0	253.3	8.6	19.0	251.1	13.4	0.72
17	0.3	300	11.3	248.5	22.2	6.4	262.5	41.9	1.33
9	0.6	300	26.1	245.7	9.4	29.9	229.3	7.7	1.00
17	0.6	300	9.6	231.6	24.0	10.0	251.2	24.9	1.59
9	0.9	300	20.4	220.7	11.1	20.3	250.3	12.8	0.90
17	0.9	300	8.9	218.9	24.9	8.6	217.7	25.6	1.59
9	0.3	150	41.8	323.7	7.6	32.6	420.4	13.1	0.76
9	0.3	600	15.9	132.0	8.2	8.8	151.7	17.7	0.77

17 (geometric mean 1.04). We do not have an established explanation. On real data (Section 8) the effect on α tracked whether the degeneracy had been setting NUTS’s trajectory length, which is a plausible reading here too but was not measured.

6.3 Random intercept only

The same simulator with $\sigma_{b1} = 0$, fitted with a random intercept only, gave seven designs: about 9, 17 and 35 visits, $N \in \{100, 300\}$, and one noisier design with $\sigma_e = 0.8$. The intercept gained 6.5–282 times and the subject-constant coefficient of age 6.6–207 times; the time slope, which has no absorbable direction when there is no random slope, moved only through the step size (0.79–3.4 times). These large ratios overstate the typical gain: in 17 of the 21 unrotated fits some parameter had $\hat{R}$ above 1.05, the intercept’s reaching 1.27, and the intercept mixed at 0.002–0.04 ESS per draw everywhere except the noisy design. Every rotated fit had $\hat{R} \leq 1.004$. The comparison is between a converged fit and one that had not converged at the standard run length, so its order of magnitude is the finding. Posterior means agreed to a median of 0.025 posterior standard deviations (maximum 0.175).

6.4 Stress grid

Section 4.4 names two regimes where the rotation should be weakest: few subjects each carrying a lot of information, and strongly correlated random effects. A stress grid targeted both, with about 35 visits crossed with $N \in \{50, 100\}$ and $\rho \in \{0.3, 0.95\}$, plus $\rho = 0.95$ at 9 visits ($N = 100$) and at 17 visits ($N = 300$). The criterion fixed before the run was that the rotation must not fall below the unrotated fit on the reported coefficients in any design. It did not (Table 2); the lowest design was 5.5 times and the lowest single seed 3.2 times.

The design with the fewest subjects and the most information per subject gained least, as the fixed-metric residual predicts. Strong correlation did not hurt: at $\rho = 0.95$ the gains were as large as

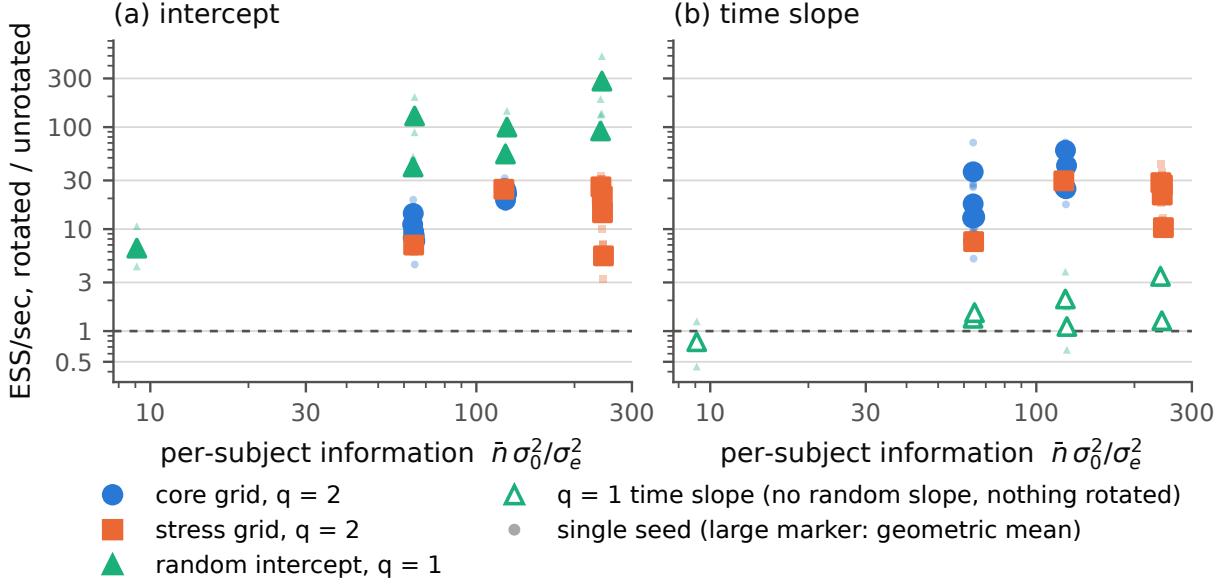


Figure 3: ESS per second of the reported coefficient, rotated over unrotated fit on the same simulated data, against the per-subject information $\bar{n} \sigma_0^2 / \sigma_e^2$ ($\sigma_0 = 0.8$). Large markers: geometric mean over three seeds in one design; small: single seeds. Core and stress grids have a random intercept and slope ($q = 2$); the $q = 1$ grid has a random intercept only. In (b), open triangles are the $q = 1$ time slope, which has no random effect and so nothing to rotate. The $q = 1$ intercept ratios compare against unrotated fits that had mostly not converged at this run length (Section 6.3), so only their order of magnitude is meaningful.

or larger than at $\rho = 0.3$, because the unrotated fit degrades faster. The predicted weakness is visible only against a variant that removes the absorbable directions from the likelihood’s coordinates and restores the reported coefficients with an exact correction (Section 9.1). That variant has no fixed-metric residual, and in the 35-visit designs it was 1.4–2 times more efficient than the rotation, while being about equal to it with 9 visits. Posterior means agreed with the unrotated fit’s to a median of 0.029 posterior standard deviations (maximum 0.166, 90 pairs), there were no divergences, and wall time was 0.86–1.06 times the unrotated fit’s.

6.5 Calibration

Proposition 3 says the rotation cannot change the posterior, so a calibration study checks the implementation rather than the method. With $N = 300$ and 100 replicates, all passing an $\hat{R} \leq 1.05$ gate on the regression parameters, coverage of 95% intervals agreed within 0.01 and the ratio of mean posterior standard deviation to the standard deviation of posterior means within 0.015 for every estimand (Table 3); the Monte Carlo standard error of a 95% coverage is about 0.011. Two features appear in both arms and belong to the model and design: slightly wide intervals for β_0 , and an upward bias of about 3 Monte Carlo standard errors (2.8%) in α . The latter is ordinary finite-sample bias. Neither the spline baseline hazard nor the simulator’s hazard calibration explained it

Table 2: Stress grid: ESS-per-second ratios, rotated over unrotated (geometric means over 3 seeds).

visits	N	ρ	intercept	time slope	α
35	50	0.30	5.5	10.3	1.60
35	50	0.95	21.1	21.6	2.42
35	100	0.30	14.5	27.1	1.58
35	100	0.95	26.0	28.6	1.73
9	100	0.95	7.0	7.5	0.74
17	300	0.95	24.6	29.7	1.09

Table 3: Calibration, $N = 300$, 100 replicates: coverage of 95% intervals, ratio of mean posterior SD to the SD of posterior means, and bias in Monte Carlo standard errors.

estimand	unrotated			rotated		
	cover	SD ratio	bias/se	cover	SD ratio	bias/se
β_0	0.99	1.164	1.68	0.99	1.150	1.55
β_{time}	0.97	1.027	-0.30	0.96	1.038	-0.32
α	0.93	0.922	2.97	0.93	0.917	2.97
σ_{b0}	0.94	0.914	0.63	0.95	0.906	0.68

(paired changes of -0.0011 and -0.0016 , standard error about 0.0025), and at $N = 1200$ it fell to 1.1%, 1.7 Monte Carlo standard errors, with coverage 0.92.

7 Beyond joint models

Nothing in Sections 3–4 depends on the survival submodel except Condition S. To check that the gains are not specific to joint models, we applied the construction, in a standalone NumPyro program that does not use `jmx`, to two classical mixed models, each with a subject-constant covariate:

- **Orthodont** (Potthoff and Roy, 1964; Pinheiro and Bates, 2000): a Gaussian linear mixed model for the distance measure of 27 children at 4 ages, $\text{distance} \sim \text{age} \times \text{male} + (1 + \text{age} \mid \text{child})$ with age centred. Sex is subject-constant, so both random-effect columns have absorbable directions $\{\mathbf{1}, \text{male}\}$, $k = 2$.
- The toenail onychomycosis trial (De Backer et al., 1998; Lesaffre and Spiessens, 2001): a logistic mixed model for 1908 binary outcomes on 294 patients, $y \sim \text{trt} \times \text{time} + (1 \mid \text{patient})$. Treatment is subject-constant, so the absorbable directions are $\{\mathbf{1}, \text{trt}\}$.

Both use the non-centred parameterization with $\mathcal{N}(0, 10^2)$ priors on β , half-normal(5) priors on the standard deviations and an LKJ(2) prior on the correlation, which is the parameterization `brms` and `rstanarm` use by default. Three arms were run on each, with 4 chains of 1000 warm-up and 1000 sampling iterations and 3 seeds: unrotated with a diagonal metric, rotated with a diagonal metric, and rotated with the dense (β, U) block. Because each run takes only a few seconds including compilation, we report ESS per 1000 gradient evaluations, which does not depend on the machine.

Table 4: Two mixed models: ESS per 1000 gradient evaluations, rotated over unrotated (geometric means over 3 seeds), with the dense block and, in brackets, with a diagonal metric only.

coefficients	Orthodont (LMM, $q = 2$)	toenail (logistic GLMM, $q = 1$)
intercept	7.1 (2.3)	5.5 (2.5)
subject-constant covariate	7.1 (2.4), male	6.7 (2.8), trt
absorbed by the random slope	2.4–2.5 (0.9), age and age $\times$ male	—
not absorbable	—	1.3–1.4 (0.9), time and trt $\times$ time
variance components	1.3–1.6 (0.9)	1.1 (0.9)

Table 4 shows the same pattern as the joint model. The intercept and the subject-constant coefficients gained 5.5–7.1 times per gradient, and the coefficients absorbed by the random slope in **Orthodont** 2.4–2.5 times. The rotation with a diagonal metric alone supplied a factor of 2.3–2.8 on the intercept and subject-constant coefficients and nothing on the slope-absorbed ones; the dense block supplied the rest, as Proposition 4 predicts. Per second, including compilation, the gains on those coefficients were 4.1–5.1. Posterior means agreed with the unrotated fit’s to within 0.15 posterior standard deviations, $\hat{R} \leq 1.011$ everywhere, and there were no divergences.

8 Real data

8.1 Rotated against unrotated

We fitted two standard joint-model datasets, both distributed with JMBayes2, twice from the same preliminary mixed-model and Cox fits and the same sampler seed, unrotated and rotated, with 4 chains of 1000 warm-up and 1000 sampling iterations and 3 seeds per dataset.

- **aids** (Abrams et al., 1994; Guo and Carlin, 2004): 467 patients, $\sqrt{\text{CD4}} \sim \text{obstime} + \text{drug} + \text{gender}$ with a random intercept and slope on **obstime**, and a hazard for death depending on drug. There are three absorbable intercept directions (constant, drug, gender) and one slope direction.
- **pb2** (Murtaugh et al., 1994): 312 patients, $\log(\text{serum bilirubin}) \sim \text{year} + \text{age}$ with a random intercept and slope on year, and a hazard for death depending on drug. There are two absorbable intercept directions (constant, age) and one slope direction.

The coefficients on absorbable directions gained 17–28 times and the time slope 2.3–4.4 times (Table 5). Every posterior mean agreed between the two fits within Monte Carlo error (largest paired difference 1.65 Monte Carlo standard errors over 19 parameter-dataset pairs). The two datasets differ in one instructive way. On **aids** the unrotated trajectories were three times longer: the degeneracy was what set NUTS’s trajectory length, and removing it made every iteration cheaper, so every parameter gained, α included, and sampling time fell by 40%. On **pb2** the trajectory length was set by something else, and the parameters the degeneracy does not involve were unchanged or slightly slower.

Table 5: Real data: ESS per second, rotated over unrotated, mean over 3 seeds (per-seed range in brackets), and sampler statistics.

	aids	pbc2
intercept	26.9 [21–34]	17.2 [12–25]
subject-constant covariates	24.8, 28.5 [21–33]	16.6 [12–23]
time slope	2.3 [2.0–2.7]	4.4 [4.0–4.6]
α	2.0 [1.8–2.4]	0.95 [0.88–1.01]
$\sigma_e, \sigma_b, \rho, \gamma$	1.4–2.4	0.82–1.28
sampling time (s), rotated / unrotated	10.8 / 17.6	12.4 / 11.7
leapfrog steps per iteration	31 / 92–111	60–63 / 63
maximum $\hat{R}$, all parameters	1.010 / 1.010	1.007 / 1.018

Table 6: ESS per second relative to JMbayer2 (same data, 3 seeds).

	aids		pbc2	
	unrotated	rotated	unrotated	rotated
intercept	0.04	0.97	0.05	0.89
subject-constant covariates	0.03–0.04	0.89–0.95	0.06	0.92
time slope	2.8	6.5	0.42	1.8
α	3.2	6.5	7.6	7.3
survival coefficient γ	1.1	2.7	2.1	1.8

With a random intercept only, the same datasets show the degeneracy more starkly. Unrotated, β_0 and the subject-constant coefficients mixed at 0.04–0.07 ESS per draw, and 91–95% of their posterior variance was explained by the absorbable directions of the random effects; every other parameter mixed near one ESS per draw. The rotation raised the affected coefficients’ ESS per second 51–53 times on **aids** and 19–20 times on **pbc2** (2 seeds each), and the slowest parameter’s ESS per draw from 0.05 to 0.88 and from 0.06 to 0.78.

8.2 A blocked sampler

JMbayer2 (Rizopoulos et al., 2024) fits the same models by Metropolis-within-Gibbs and hierarchically centres the coefficients it can, which is the classical remedy for the location degeneracy (Gelfand et al., 1995). It therefore offers two comparisons: on the centred coefficients, of the rotation with centring; on the rest, and on α in particular, of HMC with a blocked sampler (Section 5). JMbayer2 was run with 4 chains of 16,000 iterations and 4,000 burn-in, well above its defaults, and its time is the whole model-fitting call. Its ESS is the one JMbayer2 reports, computed by a spectral estimator rather than NumPyro’s, so the ratios compare two ESS estimators as well as two samplers.

Unrotated, the NUTS fit trailed JMbayer2 by a factor of 17–33 on the coefficients JMbayer2 centres (Table 6). Rotated, it is at parity on them (0.89–0.97) and ahead on the others: 1.8–6.5 times on the time slope, 1.8–2.7 on γ and 6.5–7.3 on α . The rotation thus recovers what hierarchical centring buys on the regression coefficients, while keeping the non-centred parameterization for the

random-effect variances. The lead on α needs no rotation at all; it is the blocked-sampler effect of Section 5. JMBayes2 draws 4.7–7.1 times faster per second but keeps one effective draw in 29–36 for α (Figure 2).

9 Discussion

9.1 Related work

Choosing a parameterization. That the parameterization of a hierarchical model decides how fast a sampler mixes is well established. Hierarchical centring (Gelfand et al., 1995, 1996) removes the location degeneracy by sampling the centred effects; the non-centred parameterization removes the funnel instead (Papaspiliopoulos et al., 2003, 2007); partial non-centring interpolates, and ASIS (Yu and Meng, 2011) interweaves the two within a Gibbs sampler. Gorinova et al. (2020) automate the choice per variable in probabilistic programs. All of these choose between centred and non-centred coordinates for each random effect. The rotation does not: it keeps the non-centred parameterization, with its benefits for the variances, and changes only which linear combinations of the standardized effects are sampling coordinates. In the comparison with JMBayes2, which centres the coefficients it can, this recovered what centring buys on those coefficients (Section 8). Browne et al. (2009) studied three reparameterizations for multilevel survival models under Gibbs sampling. Their orthogonal parameterization orthogonalizes the fixed-effect predictors among themselves, as Stan’s QR reparameterization does (Stan Development Team, 2025), but they observe that predictors correlated with the cluster indicator vectors make fixed and random effects “play similar roles” and suggest orthogonality to those indicators as future work. The absorbable subspace is that span, handled by rotating the random effects rather than the predictors.

Removing the directions. A second family removes the absorbable directions. Sum-to-zero constraints are the classical example. Zanella and Roberts (2021) show that imposing such constraints always improves the Gibbs sampler for crossed random effects, and that it also helps NUTS dramatically on a Poisson crossed model (minimum ESS from 3.7 to 4977). Their construction samples the posterior conditioned on the constraint, and Stan’s `sum_to_zero_vector` likewise changes the prior unless compensated. Restricted spatial regression (Reich et al., 2006; Hughes and Haran, 2013) projects spatial random effects off the fixed-effect design, and is known to affect inference on the fixed effects (Hodges and Reich, 2010; Hanks et al., 2015; Khan and Calder, 2022). An earlier version of `jmlax` did the same for the absorbable directions: it swept them out of the sampled random-effect coordinates. Without a correction this miscalibrated the intervals for the swept coefficients; with an exact correction it was about as efficient as the rotation in ordinary designs and up to twice as efficient in the most information-dense ones (Section 6.4), at the price of the correction step. The closest precedent is Vines et al. (1996), whose transformation is exact: for independent Gaussian random effects and a Gaussian prior on the grand mean, the directions that shift the grand mean against the mean of each factor are a priori independent of everything

else, appear in no likelihood term, and can be integrated out, leaving the grand mean plus the mean effects and sum-to-zero contrasts to sample. Their method covers the grand-mean direction only, reports contrasts rather than the original parameters, and leaves correlated random effects to further work; they also warn that the *ad hoc* alternative, recentring the effects inside each iteration, has an unknown effect on the stationary distribution. The rotation extends the exact approach to every absorbable direction, including the covariate-weighted ones and those of random slopes, to correlated random effects and to any prior on the fixed effects, and it returns the original parameters.

Analysing the directions. In the discussion of Zanella and Roberts (2021), Yang and Liu (2021) decompose the Gibbs sampler for a random-intercept model with covariates using the subject-level covariate means, which is the covariate-weighted absorbable subspace for the intercept column, and show that the convergence rate is governed by the correlation between β and the random effects along it. They use the decomposition to analyse Gibbs samplers, note that it breaks down for general designs, and do not treat random slopes; here the subspace is constructed exactly for every column from the design and used as sampling coordinates. In the same discussion Robert (2021) asked whether the effect of these parameterization choices had been studied for HMC, and the rejoinder (Zanella and Roberts, 2021) called that open. Lemma 1 and Proposition 4 are an analysis on the HMC side for the absorbable directions: what a diagonal metric can and cannot do, what a dense block adds, and where a fixed metric falls short. They complement general results on linear preconditioning for MCMC (Hird and Livingstone, 2025): the block the rotation creates is small enough to precondition densely because the model says where the ridge is.

Joint-model software. JMbays2 (Rizopoulos et al., 2024) uses Metropolis-within-Gibbs; the function `stan_jm` in the `rstanarm` package (Brilleman et al., 2018) uses NUTS with non-centred random effects and so, we expect, meets the same ridge, which we have not tested; INLAjoint (Rustand et al., 2024) uses integrated nested Laplace approximations and does not sample. To our knowledge, the application of the data-augmentation autocorrelation identity (Liu et al., 1994) to the association parameter, together with the measurement that places most of the shared variance in the baseline hazard, has not been reported before.

9.2 Costs and limits

Two costs were measured and left as they are. With about 9 visits per subject, the rotation gave α 0.72–1.00 of the unrotated fit’s ESS per second, though it gained with denser follow-up and α remained far ahead of the blocked sampler either way; the mechanism is not established. And in the most information-dense stress designs, the swept variant was up to twice as efficient, as the fixed-metric residual of Section 4.4 predicts, although the rotation was still 5–30 times ahead of the unrotated fit there. The simulation evidence comes from one design family with a spline baseline hazard. The basis construction’s rank decision is a tolerance judgement when absorbable

and identified directions are nearly collinear, although every returned direction is verified before use. Condition S is checked at each fit rather than assumed, and can fail for very flexible time bases with few observations per subject.

9.3 Extensions

A metric that follows the sampled σ_0 , for example by rescaling the dense block with the current value of L , is the obvious way to remove the residual of Section 4.4, and has not been tried. The construction needs only the fixed- and random-effects designs, so it applies to any model whose random effects enter through a linear predictor, including the GLMMs of brms and rstanarm, and could be offered there as a transformation of the non-centred random effects. Crossed random effects need one basis per grouping factor, each rotating that factor’s own index; the argument of Proposition 3 carries over unchanged.

9.4 Conclusion

The location degeneracy between fixed and random effects is shared by every hierarchical regression with subject-constant covariates, and it recurs for the time slope in every model with a random slope. It slows HMC on the coefficients an analyst reports. Rotating the standardized random effects so that the absorbable directions become a small block of coordinates, and giving that block and the fixed effects one dense metric block, removes it without changing the model: in simulation by factors of 14 and 21 for the intercept and time slope, on real joint-model data by 17–28 on the intercept and subject-constant covariates, and on the same coefficients by 5–7 per gradient in two ordinary mixed models. The association parameter of a joint model is a separate matter. Its posterior variance is largely shared with the level of the baseline hazard, and a sampler that updates it conditionally pays for that sharing in autocorrelation, whereas HMC pays, if at all, in integration steps.

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A Proofs

Proof of Proposition 2. ($\supseteq$) Let $d \in \ker M_c$ and $v = v(d)$ as in Condition S. For $i \in S_c$, Proposition 1(b) gives $X_i d + v_i z_i = 0$; for $i \notin S_c$, $d \in \ker M_c$ gives $X_i d = 0$ and $z_i = 0$, so (L) holds. Condition S gives (T). Hence $v \in \mathcal{A}_c$.

($\subseteq$) Let $v \in \mathcal{A}_c$ with witness d . By (L) and Proposition 1(b), $d \in \ker M_c$ and $v_i = -z_i^\top X_i d / z_i^\top z_i$ for $i \in S_c$. For $i \notin S_c$, (T) at any t with $z_i(t) \neq 0$ gives $v_i = -x_i(t)^\top d / z_i(t)$, which by Condition S is $v_i(d)$. So $v = v(d)$.

If $z_i(t) = 0$ for every t as well as at every observation, then column c of subject i 's random effects does not enter the likelihood at all, v_i is unrestricted, and the corresponding coordinate is identified by its prior alone; it plays no part in the ridge. $\square$

A sufficient condition for Condition S. Suppose each column of the fixed-effects design and column c of the random-effects design are, for subject i , functions $f_j(t; w_i)$ and $g(t; w_i)$ of time and baseline covariates, and that for each d and scalar s the function $\sum_j d_j f_j(\cdot; w_i) + s g(\cdot; w_i)$ lies in a linear space of functions that is determined by its values at subject i 's n_i observation times (for example, polynomials of degree less than n_i). Then, for $i \in S_c$, (L) at the observation times implies the identity at all t , which is Condition S for that subject. Subject-constant columns, a linear time column and covariate-by-time interactions satisfy this with $n_i \geq 2$.

B Reproducibility

All code is in the `jmjax` repository. The development scripts below reproduce each result; paths are relative to the package root.

result	script (output)
Figure 1, Section 4.4 numbers	dev/theory_rotation_toy.py; dev/figures/make_figures.py
Section 5, Gaussian check	dev/theory_alpha_toy.py
Section 5, R^2 and its split	dev/alpha_missing_information.R
Figure 2	dev/figures/make_figures.py
Table 1	dev/pilot_rotate_grid.R (dev/pilot_rotate_grid.csv)
Section 6.3	ROT_GRID=q1 dev/pilot_rotate_grid.R (dev/pilot_rotate_q1.csv)
Table 2	ROT_GRID=stress dev/pilot_rotate_grid.R (dev/pilot_rotate_stress.csv)
Figure 3	dev/figures/make_figures.py
Table 3	dev/study_calibration.R
α bias, $N = 1200$	dev/study_alpha_bias.R (dev/alpha_bias_results.csv)
Table 4	dev/generality/mixed_models.py (dev/generality/results_orthodont_toenail.csv)
Tables 5 and 6	dev/study_realdata_rotate.R
Section 8, random intercept only	dev/pilot_q1_intercept.R
α against JMbates2, 1–12 covariates	dev/study_orth_vs_jmbates2.R
Example 1, basis checks	dev/verify_absorbable_basis.py